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**STM32 Based Automatic Cooling System
with Multi Level Temperature
Monitoring**

EE5203 MICROPROCESSORS
SEMESTER PROJECT

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ABSTRACT

In this project, a temperature-controlled automatic cooling and warning system is designed and implemented using an STM32 microcontroller. The system monitors temperature continuously through an LM35 temperature sensor, and the obtained analog signal is processed using the built-in analog-to-digital converter (ADC) of the microcontroller. Based on predefined temperature thresholds defined in the software, the system operates in three different modes. Under normal temperature conditions, both the cooling fan and the buzzer remain inactive. During heating conditions, the fan operates at 75% duty cycle using PWM control and an audible warning is generated by a buzzer. In the case of overheating, the fan operates at full speed (100% duty cycle) and the buzzer remains active to indicate a critical temperature level. The system utilizes fundamental microprocessor concepts such as ADC-based sensor interfacing, GPIO control, timer-based periodic measurement, PWM generation, and decision-making algorithms. The results demonstrate that the proposed STM32-based system provides reliable temperature monitoring, effective cooling, and enhanced safety through multi-level visual and audible feedback.

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1. INTRODUCTION

Temperature monitoring and control are essential requirements in many electronic and electromechanical systems to ensure safe operation, performance stability, and long-term reliability. Excessive temperature levels can lead to system malfunction, accelerated component aging, or permanent damage. For this reason, automatic cooling and warning mechanisms are widely employed to prevent overheating and to provide timely feedback before critical temperature conditions occur.

With the rapid advancement of embedded systems, microcontroller-based solutions have become a practical and efficient approach for implementing temperature control applications. Modern microcontrollers integrate peripherals such as analog-to-digital converters (ADC), general-purpose input/output (GPIO) ports, timers, pulse-width modulation (PWM) units, and interrupt systems. These features enable real-time data acquisition, signal processing, and precise control of external devices within a compact and cost-effective platform. Among various microcontroller families, STM32 devices are commonly used in academic and industrial environments due to their high performance, low power consumption, and rich peripheral support.

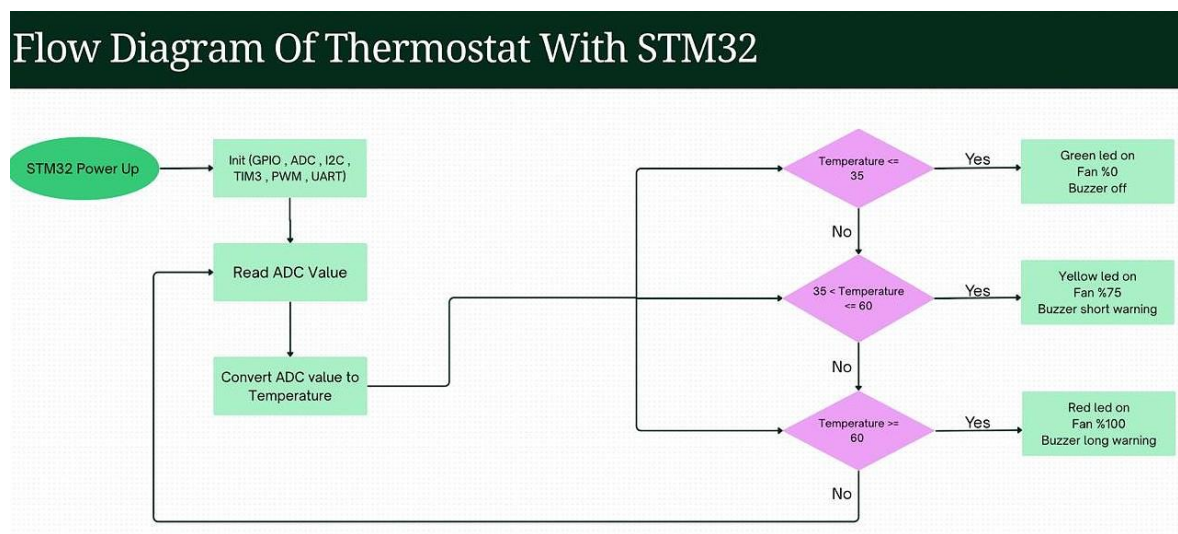
In this project, a temperature-controlled automatic cooling and warning system based on an STM32 microcontroller is designed and implemented. The system continuously monitors temperature using an LM35 temperature sensor, which provides an analog voltage proportional to the measured temperature. The STM32 microcontroller digitizes this signal using its ADC unit and evaluates the temperature according to predefined threshold values defined in the software. Based on the measured temperature, the system operates in three different modes. Under normal temperature conditions, both the cooling fan and the buzzer remain inactive. During heating conditions, the fan operates at 75% duty cycle using PWM control and an audible warning is generated by a buzzer. In the case of overheating, the fan operates at full speed 100% duty cycle, and the buzzer remains active to indicate a critical temperature level.

This project aims to demonstrate the practical application of fundamental microprocessor concepts covered in the course, including ADC-based sensor interfacing, GPIO control, timer-based periodic measurement, PWM generation, and decision-making algorithms. By integrating cooling and warning mechanisms within a single embedded system, the proposed design provides an effective, reliable, and scalable solution for automatic temperature monitoring and control applications.

2. METHODS

The methodology section describes the procedures followed during the design and implementation of the temperature-controlled cooling and warning system. This section explains how temperature data is collected and analyzed, the control methods applied, and the hardware software design approach adopted to realize the system.

Table 1. Flow Diagram of Thermostat with STM32



2.1. Data Collection and Analysis

Temperature data is collected using an LM35 temperature sensor, which generates an analog output voltage linearly proportional to the measured temperature with a sensitivity of 10 mV/°C. The sensor output is connected to an analog input channel of the STM32 microcontroller and sampled using the internal analog-to-digital converter (ADC).

The ADC converts the analog sensor voltage into a digital value based on the reference voltage and resolution of the microcontroller. The obtained ADC value is processed in software to calculate the corresponding temperature in degrees Celsius. Temperature measurements are taken periodically using a timer-based mechanism to ensure consistent and stable data acquisition. The calculated temperature values are continuously monitored and analyzed in real time to determine the operating state of the cooling and warning system.

2.2. Applied Methods

An embedded systems-based experimental method is applied in this project. Since the system operates in real time, statistical or historical analysis methods are not required. Instead, a deterministic threshold-based control method is implemented in software.

In this approach, the measured temperature is compared with predefined threshold values defined in the program. Based on this comparison, control decisions are made to regulate the fan speed and buzzer activation. This method is chosen due to its simplicity, reliability, and suitability for real-time microcontroller applications.

2.2.1. Design Constraints

The design of the system is subject to several hardware and software constraints. One of the main hardware constraints is the current limitation of the STM32 GPIO pins, which are insufficient to directly drive a 5V DC fan. To overcome this limitation, a switching element such as a transistor or MOSFET is used to control the fan safely. Additionally, a flyback diode is included to protect the circuit against voltage spikes caused by the inductive nature of the fan.

Another constraint is related to measurement accuracy. The resolution of the ADC and the reference voltage directly affect the precision of temperature calculations. Noise and small voltage fluctuations may influence sensor readings; therefore, stable sampling intervals and proper grounding are considered during the design.

From a software perspective, the control algorithm must operate within limited processing time and memory resources. For this reason, a simple threshold-based algorithm is preferred to ensure fast execution and predictable behavior.

2.2.1.1. Hardware Constraints

From a hardware perspective, the main limitation arises from power and current handling capabilities. The STM32 GPIO pins cannot directly drive the 5V DC fan due to insufficient current output. To overcome this limitation, a transistor or MOSFET is used as a switching element.

Another hardware constraint is related to sensor placement and wiring. Improper placement of the LM35 sensor or long wiring connections may introduce noise and affect measurement accuracy. Therefore, careful sensor positioning and short signal paths are considered during the hardware design.

2.2.1.2. Software Constraints

The software design is constrained by real-time operation requirements and limited computational resources. The temperature measurement and fan control algorithm must execute within a short and predictable time frame to ensure timely response to temperature changes. Additionally, the ADC sampling rate and timer configuration must be selected carefully to balance responsiveness and system stability.

Memory usage and code simplicity are also considered as constraints. The control algorithm is intentionally kept simple and threshold-based to ensure reliability and ease of debugging, while still fulfilling the functional requirements of the system.

2.2.2. Measurements Setups

The measurement setup consists of the LM35 temperature sensor, STM32 microcontroller, 5V DC fan, buzzer, and associated interface circuitry. The system is powered using a regulated power supply to ensure stable operation during testing.

Temperature measurements are performed by periodically sampling the LM35 output through the ADC. The system behavior is observed under three different temperature conditions: normal temperature, heating condition, and overheating condition. During normal operation, both the fan and buzzer remain inactive. In heating conditions, the fan operates at 75% duty cycle using PWM control and the buzzer is activated. In overheating conditions, the fan operates at full speed (100% duty cycle) and the buzzer remains active. The responses of the fan and buzzer are observed and recorded to evaluate system performance.

Table 2. Equipments of experiment

Experiment Equipments	Purpose
STM32 NUCLEO-F401RE	Main microcontroller used for data processing, control logic, and peripheral management
LM 35	Measuring ambient temperature and providing an analog voltage proportional to temperature
Led (Green , Yellow , Red)	Visual indication of system states (normal, heating and overheating)
Buzzer	Providing audible warning during heating and overheating conditions
Resistor	Current limiting and signal conditioning for LEDs and sensor circuits
DC Fan	Cooling the system under heating and overheating conditions
Breadboard	Prototyping and assembling the electronic circuit without soldering
Connection Cables	Establishing electrical connections between components
Transistor	Driving the DC fan safely by amplifying the control signal from the microcontroller
SSD 1306 OLED Screen with I2C	Monitoring fan speed and sensor temperature

3. IMPLEMENTATION AND RESULTS

This section presents the implementation of the proposed temperature-controlled cooling and warning system and discusses the experimental results obtained during testing. The system was implemented using the selected hardware components and programmed according to the methodology described in the previous section. The performance of the system was evaluated by observing its response under different temperature conditions.

3.1. Designed System

The implemented system consists of an STM32 microcontroller, an LM35 temperature sensor, a 5V DC cooling fan, a buzzer, and the required interface circuitry. The LM35 sensor continuously monitors temperature and provides an analog voltage proportional to the measured temperature. This signal is sampled by the ADC unit of the STM32 microcontroller and processed in real time.

The fan is controlled using pulse-width modulation (PWM) to enable different operating speeds depending on the temperature level. Since the microcontroller GPIO pins cannot directly supply the required current for the fan, a switching element is used to safely drive the fan. The buzzer is controlled via a GPIO output and provides audible feedback during abnormal temperature conditions.

The system operates in three distinct temperature states. Under normal temperature conditions, both the fan and the buzzer remain inactive. When the temperature reaches the heating condition, the fan operates at 75% duty cycle and the buzzer is activated to provide a warning. In the case of overheating, the fan operates at full speed (100% duty cycle), and the buzzer remains active to indicate a critical temperature level.

3.2. Collected Data

During the experimental phase, temperature data was collected using the LM35 sensor and processed by the STM32 microcontroller. The collected data consists of temperature readings along with the corresponding fan speed and buzzer status.

Measurements were taken periodically using a timer-based sampling mechanism. The system behavior was observed under normal temperature, heating, and overheating conditions. The collected data confirms that the temperature values were read consistently and that the system responded correctly to changes in temperature.

Table 3. Working Principles of fan speed and buzzer

Temperature (°C)	Fan Speed	Buzzer
Normal	Off	Off
Heating	%75 speed	On
Overheating	%100 speed	On

3.3. Comparisons

The experimental results were compared with the expected behavior defined during the system design phase. According to the control logic, the fan should remain off under normal temperature conditions, operate at 75% duty cycle during heating, and operate at full speed during overheating. Similarly, the buzzer should activate only during heating and overheating conditions.

The comparison between theoretical expectations and experimental observations shows strong agreement. In all tested scenarios, the system responded as intended. Minor response delays were observed due to ADC sampling intervals and PWM update timing; however, these delays are acceptable and do not affect overall system performance.

3.4. Results

The results demonstrate that the STM32-based temperature-controlled cooling and warning system operates reliably and effectively. The system successfully distinguishes between different temperature levels and applies appropriate control actions to the fan and buzzer.

The use of PWM enables efficient fan speed control, while the buzzer provides clear audible warnings during abnormal temperature conditions. Overall, the implementation validates the proposed design and confirms that the system meets the project objectives by providing accurate temperature monitoring, effective cooling, and enhanced operational safety.

4. CONCLUSION

This project focused on the design and implementation of an STM32-based temperature-controlled cooling and warning system. The main objective was to develop an automatic system capable of monitoring temperature in real time and responding appropriately to different temperature conditions in order to enhance operational safety and reliability.

Throughout the study, an LM35 temperature sensor was used to acquire temperature data, which was processed by the ADC unit of the STM32 microcontroller. Based on predefined temperature thresholds, the system was designed to operate in three modes: normal temperature, heating condition, and overheating condition. Under normal conditions, both the cooling fan and the buzzer remained inactive. During heating conditions, the fan operated at 75% duty cycle using PWM control and an audible warning was generated by the buzzer. In overheating conditions, the fan operated at full speed and the buzzer remained active, providing a clear indication of a critical temperature level.

The problem statement defined in the introduction, which aimed to achieve automatic temperature monitoring, effective cooling, and early warning without human intervention, was successfully addressed. The implemented system demonstrated stable temperature measurement, correct multi-level control of the fan, and reliable activation of the buzzer according to the defined temperature ranges.

The results obtained from the experimental tests confirm that the system responds consistently and predictably to temperature changes. The fan speed transitions between off, 75%, and 100% duty cycles occurred as expected, and the buzzer provided timely audible alerts during heating and overheating conditions. These outcomes indicate that the proposed design meets the functional requirements of the project.

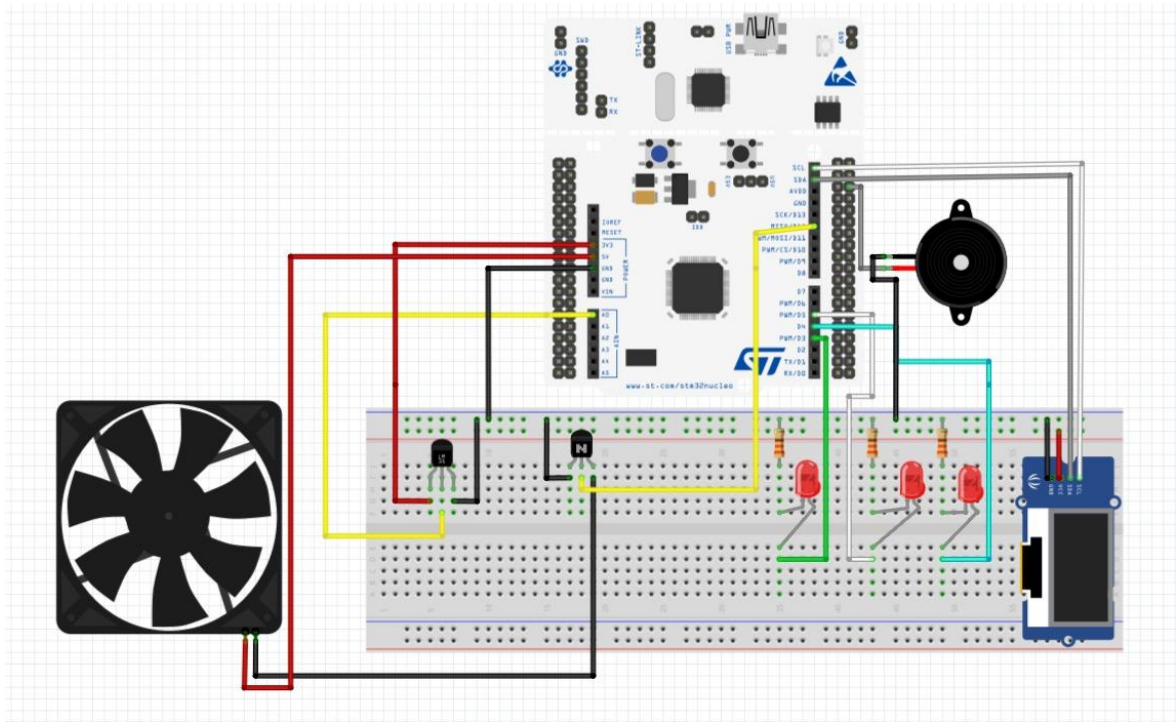
As a recommendation for future work, the system can be further improved by adding a display module to show real-time temperature values, implementing UART communication for data logging, or introducing adaptive control algorithms to dynamically adjust the temperature thresholds. Such enhancements would increase the functionality and applicability of the system in more advanced temperature control applications.

REFERENCES

- STMicroelectronics. (2023). *STM32F401xB/C and STM32F401xD/E advanced Arm®-based 32-bit MCUs reference manual*. STMicroelectronics.
- STMicroelectronics. (2023). *STM32 NUCLEO-F401RE user manual*. STMicroelectronics.
- Texas Instruments. (2016). *LM35 precision centigrade temperature sensors datasheet*. Texas Instruments.
- Doebelin, E. O. (1985). *Control system principles and design*. New York: John Wiley & Sons.
- Mazidi, M. A., Mazidi, J. G., & McKinlay, R. D. (2014). *The ARM microcontroller and embedded systems: Using assembly and C language*. Pearson Education.
- Kumar, S., & Singh, M. (2016). Design and implementation of temperature controlled fan using microcontroller. *International Journal of Engineering Research and Applications*, 6(4), 45–49.
- Monk, S. (2018). *Programming embedded systems with C and GNU development tools*. McGraw-Hill Education.

APPENDIX A

Project Schematic at Fritzing Program



GitHub Link of the Project:

<https://github.com/ef3ocak/Thermostat>

Youtube Link of the Project:

<https://youtu.be/7Jzj7MmtR-4?si=92AjI05WjcAm7sXZ>